01

Background & Introduction

What are Generative Models?

A generative model learns the underlying distribution of data and can generate new samples from it.

Training data~$p_{data}(x)$

Learning →

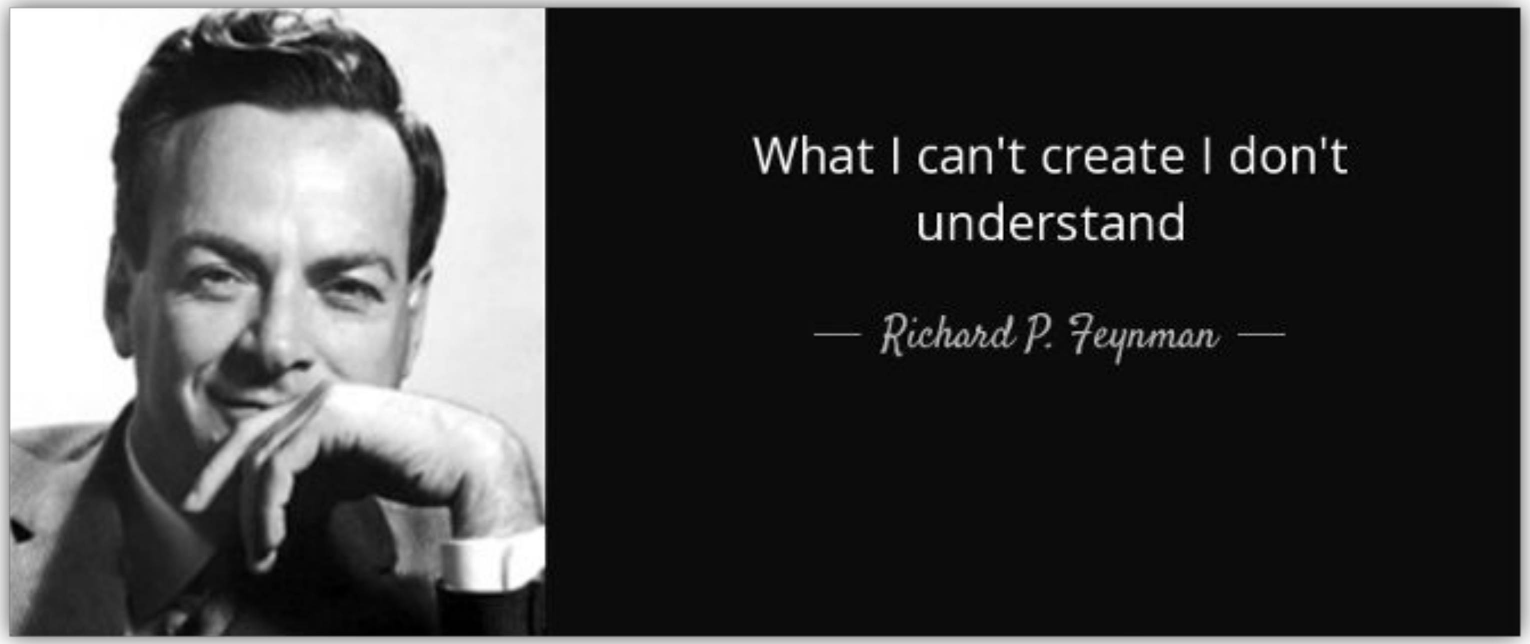
$p_\theta(x)$

Data generation distribution

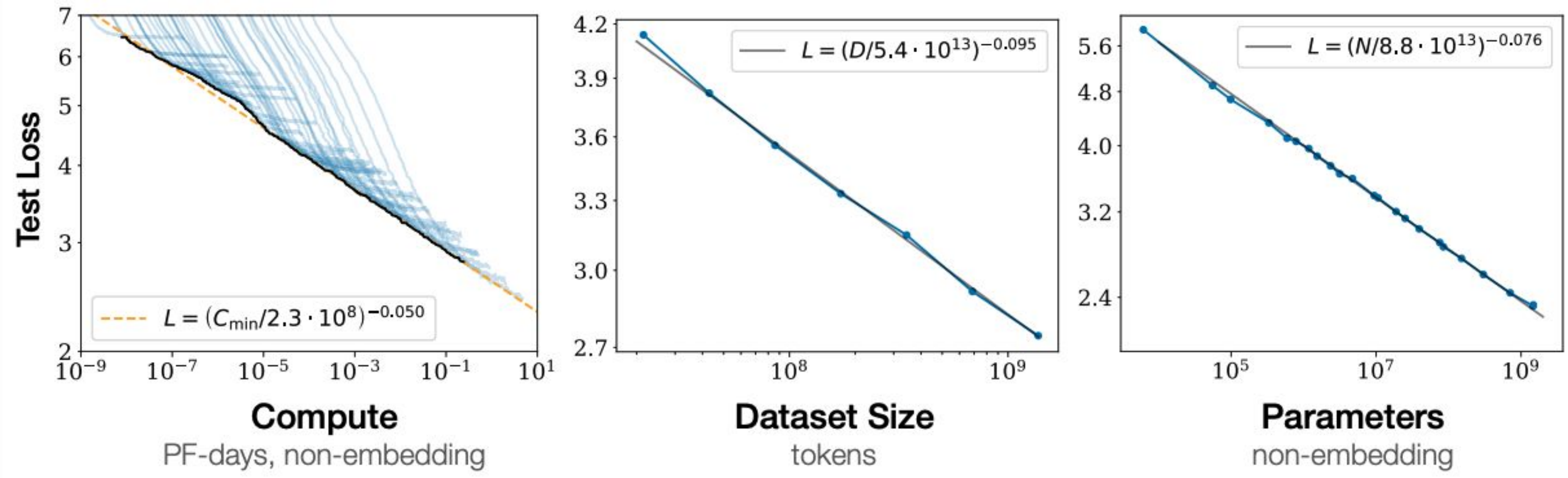
Generating →

New samples

Why “Generative” Models?



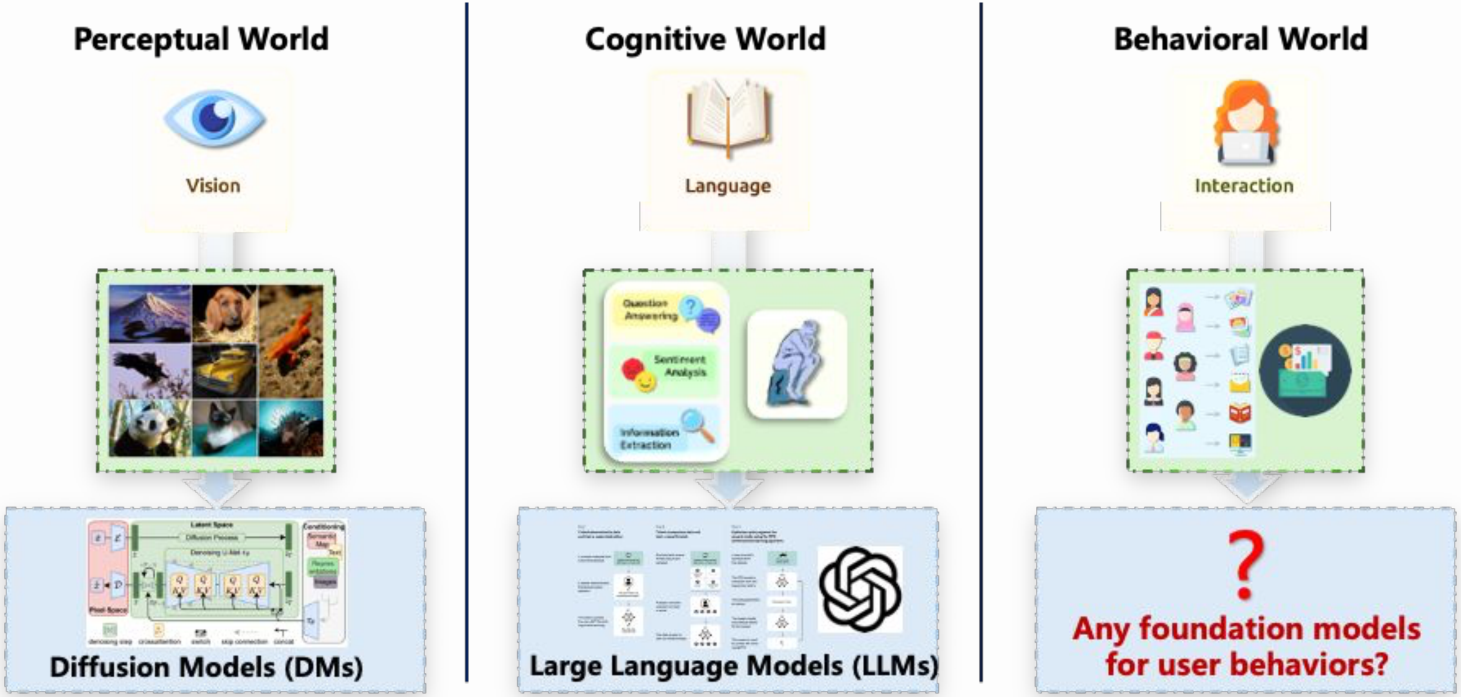
Scaling Law as a Pathway towards AGI



Scaling laws provide a framework for understanding how **model size**, **data volume**, and **test-time computing** might lead to advanced AI capabilities.

Understanding Scaling Laws for Recommendation Models. Arxiv 2022.

As the Reflection of Real World



Where are We Now?

In language and vision:

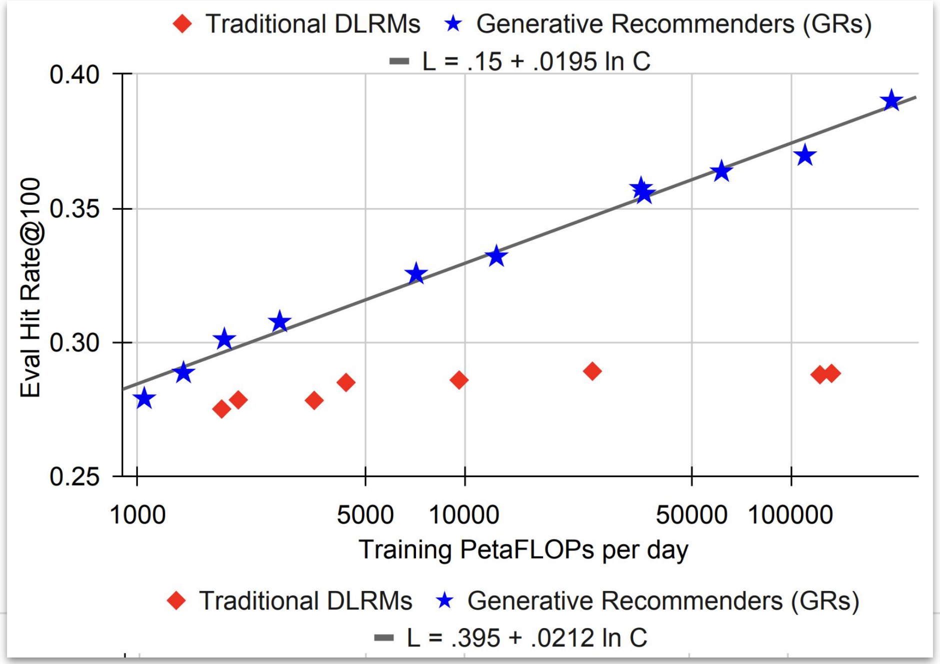
- Generative models are dominating.
- Scaling law has been witnessed.
- Large language/diffusion models have been established.

In recommendation:

- Initial attempts on building large generative recommendation models

Why “Generative” Recommendation?

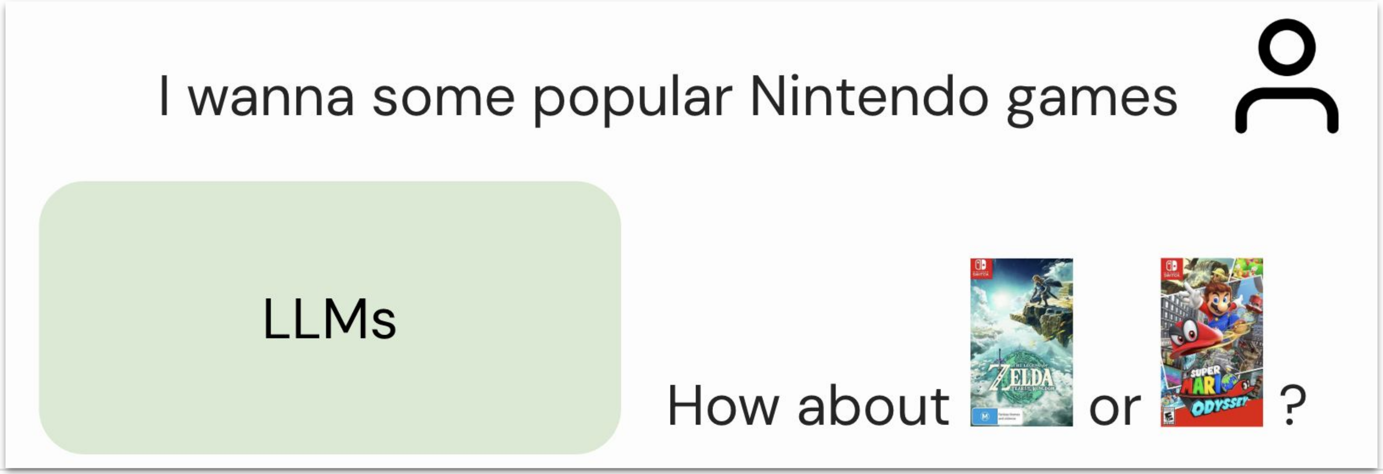
(1) Better scaling behavior



Zhai et al. Actions Speak Louder than Words: Trillion-Parameter Sequential Transducers for Generative Recommendations. ICML 2024.

Why “Generative” Recommendation?

(1) Better scaling behavior
(2) Better alignment with other modalities (text, image, audio, …);

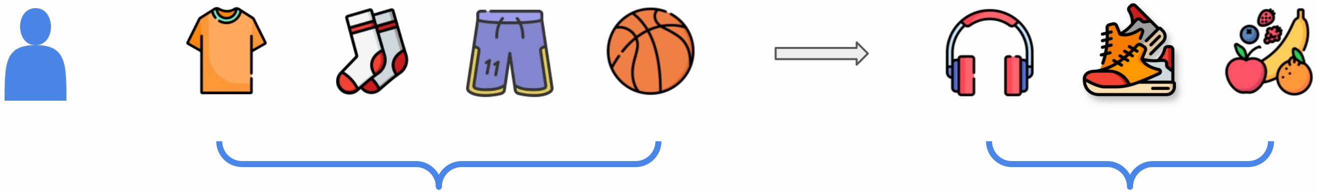


Language Modeling vs. Recommendation

Language Modeling

Will be the 46th film overall in the → Marvel Cinematic Universe

Recommendation



User Historical Actions Future Actions

However ...

Language Modeling

- Dense world knowledge
- Text tokens (10^5)

Recommendation

- Sparse user-item interactions
- Items (10^9)

Scaling laws rarely apply to traditional recommendation models.

What is “Tokenization”?

Convert

human-readable data — Text, Image, Action, …

Into

machine-readble formats — Sequence of Tokens

Language Tokenization

Human-readable Data 👨 👩

Will be the 46th film overall in the

❓ Machine cannot directly read

Language Tokenization

Human-readable Data 👨👩

Will be the 46th film overall in the

Machine-readable Data 🤖

Tokenize

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262

a token

Language Tokenization

Human-readable Data 👨👩

Will be the 46th film overall in the

Tokenize

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262 9923, 21932, 1512, 11950

token-by-token generation

Language Tokenization

Human-readable Data 👨 👩

Will be the 46th film overall in the

Tokenize

Machine-readable Data 🤖

8743, 307, 262, 6337, 400, 2646, 4045, 287, 262 9923, 21932, 1512, 11950

Detokenize

Marvel Cinematic Universe

Action Tokenization

Human-readable Data 👨 👩

Machine-readable Data 🤖

Tokenize

?

Action Tokenization

Human-readable Data 👨 👩

👕 🧦 🩳 🏀

Machine-readable Data 🤖

↓ **Tokenize**

i3487, i124, i19240, i772

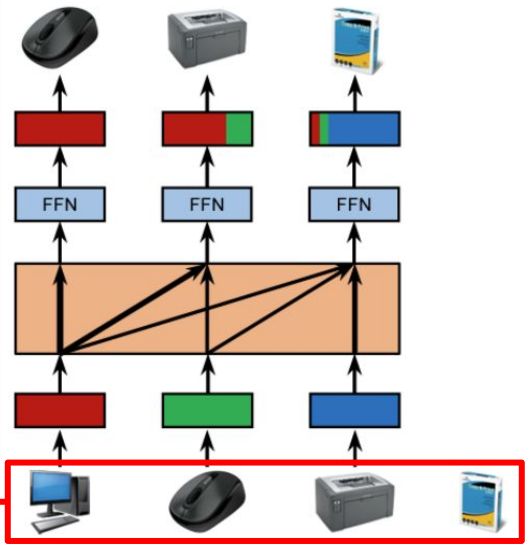
One token per action? 🤔

Action Tokenization

One token per action

Example: SASRec [ICDM'18]

Each item is indexed by a unique **item ID**, corresponding to a **learnable embedding**



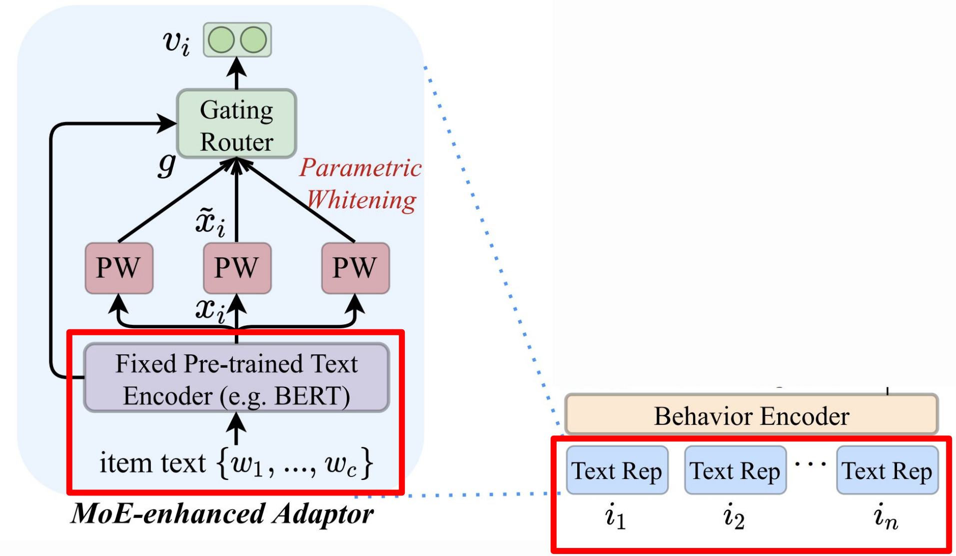
Kang and McAuley. Self-Attentive Sequential Recommendation. ICDM 2018.

Action Tokenization

One token per action

Example: UniSRec [KDD'22]

Each item is indexed by a unique **item ID**, corresponding to a **fixed representation**

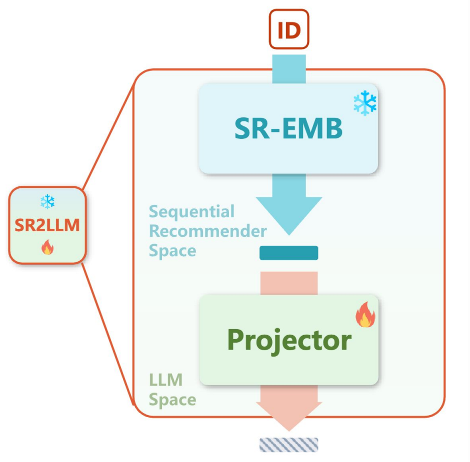


Hou et al. Towards Universal Sequence Representation Learning for Recommender Systems. KDD 2022.

Action Tokenization

One token per action

Example: LLaRA [SIGIR'24]



Align **item representations** with text tokens in LLMs

(b) Hybrid prompting method.

Input: This user has watched Titanic $[\mathrm{emb}_s^{14}]$, Roman Holiday $[\mathrm{emb}_s^{20}]$, ... Gone with the wind $[\mathrm{emb}_s^{37}]$ in the previous. Please predict the next movie this user will watch. The movie title candidates are The Wizard of Oz $[\mathrm{emb}_s^{5}]$, Braveheart $[\mathrm{emb}_s^{42}]$,..., Waterloo Bridge $[\mathrm{emb}_s^{20}]$,... Batman & Robin $[\mathrm{emb}_s^{19}]$. Choose only one movie from the candidates. The answer is

Output: Waterloo Bridge.

Liao et al. LLaRA: Large Language-Recommendation Assistant. SIGIR 2024.

Action Tokenization

One token per action

How many tokens in LLMs?

Meta	Llama 3	~128,000
OpenAI	GPT-5	~200,000
Google DeepMind	Gemma 2	~256,000

Action Tokenization

One token per action

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How many item IDs?

Amazon-Reviews-2023	~48,200,000

https://amazon-reviews-2023.github.io/

Action Tokenization

One token per action

How many tokens/item IDs in LLMs/RecSys?

Having too many tokens may make the data too sparse to effectively train a large generative model

~128,000

~200,000

~256,000

~48,200,000

Action Tokenization

Human-readable Data 👨 👩

Machine-readable Data 🤖

Tokenize

Can we tokenize the action data into a distribution that generative models can easily fit?

Action Tokenization

Human-readable Data 👨 👩

👕 🧦 🩳 🏀

⇩ **Tokenize**

Machine-readable Data 🤖

- **Text Descriptions** (LLM-based Generative Rec)
- **Semantic IDs** (SemID-based Generative Rec)

Schedule Overview

Time	Session
13:45 - 14:15	Part 1: Background and Introduction
14:15 - 15:00	Part 2: LLM-based Generative Recommendation
15:00 - 15:30	Part 3: Semantic IDs
15:30 - 16:00	Coffee Break
16:00 - 17:15	Part 4: Semantic ID-based Generative Recommendation
17:15 - 17:45	Part 5: Open Challenges and Beyond
17:45 - 18:00	Q&A Discussion

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Takeaway of Intro

(1) Why generative rec?

Scaling law; Alignment with other modalities

Takeaway of Intro

(1) Why generative rec?

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(2) Tokenization

Human-readable data -> Machine-readable data

Takeaway of Intro

(1) Why generative rec?

Scaling law; Alignment with other modalities;

(2) Tokenization

Human-readable data -> machine-readable data

(3) Action Tokenization

One token per action -> too sparse for generative models;

Text or semantic IDs?